[CO-1]

Assessment Tool-01

Name - M.Samanvi

Rg.No - 192425318

slot - B

code - MLA0401

47/50

94% marks

course Name - Deep Learning

1. A coin is tossed 10 times & the outcome are

HHTHHH THHT

Definition -

MLE is a statistical method used to estimate the value of unknown Parameter from data.

Formula-

$$\hat{p} = \frac{\text{Number of successes}}{\text{Total Number of Trials}}$$

Total Tosses = 10

Number of Heads = 7

calculation-

$$\hat{p} = \frac{7}{10}$$

$$= 0.7$$

Answer

$$\boxed{P = 0.7}$$

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2. out of 50 student , 42 Passed the exam

MLE estimats the Probabilly of success by dividing the number of successful

Formula -

$$\hat{p} = \frac{\text{Number of successes}}{\text{Total Observations}}$$

Given,

Total student = 50

Passed student = 42

calculation -

$$\hat{p} = \frac{42}{50}$$

$$\hat{p} = 0.84$$

Answer -

$\boxed{0.84}$

conclusion -

student Passes the exam is 84%

MLE estimate the Probability of an event using the observed sample data.

Formula

$$\hat{p} = \frac{\text{Number of Defective Bulbs}}{\text{Total Bulbs}}$$

Given,

Total Bulbs = 100

Bulbs = 8

calculation -

$$\hat{p} = \frac{8}{100}$$

$$\hat{p} = 0.08$$

Answer

The estimated Probability that a bulb is defective is 0.08 (8%)

4.

MLE estimate the Probability of an event by dividing the number of successful event by the total number of observation

Formula -

$$\hat{p} = \frac{\text{Number of Success}}{\text{Total Trials}}$$

Given,

60 = Total Rolls

18 = Number of sixes

calculation

$$\frac{18}{60}$$

0.3

Probability of getting 6 is 0.3 (30%)

5.

Given,

$X = 4$

$W = 2$

Actual output $= 10$

$\eta = 0.01$

Gradient $= 4$

i) Predicted output

$y = \omega \times x$

$y = 2 \times 4$

$\boxed{y = 8}$

ii) error

error = Actual - Predicted output

$= 10 - 8$

$\boxed{= 2}$

iii) $New_{weight} = \omega - (\eta \times \text{Gradient})$

$= 2 - (0.01 \times 4)$

$\boxed{= 1.96}$